CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest number divisible by each of 9, 12 and 18 is: 1

(a) 45 **(b)** 37 **(c)** 35 **(d)** 36

2. The greatest number that divides 84, 126 and 210 leaving no remainder is: 1

(a) 42 **(b)** 44 **(c)** 43 **(d)** 41

3. For the 3-digit number 3d7, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 17

4. Which is the greatest 3-digit number that leaves remainder 5 when divided by 11? 1

(a) 1006 **(b)** 995 **(c)** 996 **(d)** 984

5. The sum of the first 5 prime numbers after 10 is: 1

(a) 82 **(b)** 85 **(c)** 83 **(d)** 84

6. The largest prime number less than 200 is: 1

(a) 200 **(b)** 198 **(c)** 199 **(d)** 201

7. The product of the two smallest prime factors of 910 is: 1

(a) 35 **(b)** 14 **(c)** 10 **(d)** 7

8. The smallest 4-digit number divisible by 23 is: 1

(a) 1012 **(b)** 1035 **(c)** 1013 **(d)** 989

9. **Assertion (A):** 15 and 28 are co-prime numbers. 1
Reason (R): 28 is a composite number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** If a number is divisible by 9, it is divisible by 3. 1
Reason (R): 3 is a factor of 9.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the prime factors of 90 and the smallest number to multiply to make it divisible by 77. 2

12. Find the smallest number divisible by 12, 16 and 24. 2

13. Show that the product of 5 consecutive numbers starting from 47 is divisible by 5. 2

14. Find the prime factors of 105 and the smallest number to multiply to make it divisible by 44. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Three traffic lights change every 24, 40 and 60 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

16. Find the greatest number that divides 215 and 248 leaving remainder 6 in each case. 3

17. Find the smallest number which when divided by 6, 9 and 15 leaves remainder 2 each time. 3

OR

Three traffic lights change every 45, 75 and 90 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

18. Find the greatest number that divides 113 and 155 leaving remainder 8 in each case. 3

OR

Find the smallest number which when divided by 14, 21 and 28 leaves remainder 6 each time. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sana is using the Sieve of Eratosthenes on the numbers from 1 to 30. She crosses out 1 and all multiples of 2, 3 and 5 except the numbers 2, 3 and 5 themselves. 5

(i) How many prime numbers are there from 1 to 30?
(ii) Write the primes between 20 and 30.
(iii) Is 27 a prime? Give a reason.
(iv) Use the divisibility test for 3 to show that 57 is not a prime.

20. A rectangular garden is 108 m by 84 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,
(ii) the number of beds,
(iii) the cost of fencing all beds at ₹5 per metre if each bed is fenced separately.

OR

Find the smallest 4-digit number divisible by 18, 24 and 30. Also find the largest 4-digit number divisible by all three. 5

END OF QUESTION PAPER — Total: 40 Marks